

CC-CV Charging of Lithium-ion Battery for Electric Vehicle Applications

Biranchi Narayan Kar¹
Dept. of Electrical Engineering
¹Seemanta Engineering College
Mayurbhanj, Odisha, India
biranchinarayan.kar@gmail.com

Vineet Sharma²
Dept. of Electrical Engineering
²Poornima College of Engineering
Jaipur, India
vineet.sharma@mnit.ac.in

Raghvendra Tiwari³
Dept. of Electrical Engineering
³Theerthankar Mahavir University
Moradabad, India,
raghav.rni@gmail.com

Abstract— In the realm of electric vehicles (EVs), efficient and reliable charging methods are fundamental to advancing the widespread adoption and sustainability of electric transportation. Among the various charging techniques, constant current constant voltage (CC-CV) charging has emerged as a prominent method for replenishing energy in lithium-ion batteries, the backbone of EV propulsion systems. CC-CV charging combines the benefits of both constant current (CC) and constant voltage (CV) charging phases to deliver a controlled and optimized charging process. During the CC phase, a constant charging current is supplied to the battery, ensuring a steady flow of electrical energy. As the battery approaches full capacity, the charging process transitions to the CV phase, where the charging voltage is held constant to prevent overcharging. This dual-phase charging strategy offers several advantages, including faster charging times, improved energy efficiency, and enhanced battery lifespan. This paper explores the principles, modelling, and MATLAB simulation of CCCV charging for lithium-ion battery.

Keywords—Lithium-ion battery, PI Controller, CC-CV charging

I. INTRODUCTION

The Constant Current-Constant Voltage (CC-CV) charging method is a pivotal technique for efficiently charging lithium-ion batteries, particularly in the realm of electric vehicles (EVs)[1]. This dual-phase approach begins with the application of a constant current to rapidly charge the battery until it reaches a predefined voltage level [2], thereby maximizing charging speed and ensuring optimal energy transfer. Upon reaching the designated voltage threshold, the charging method transitions to a constant voltage phase, where the current is gradually reduced as the battery approaches its full charge state. This careful management of charging conditions is essential not only for enhancing battery life but also for improving overall efficiency, crucial for the performance and sustainability of electric vehicles in the competitive automotive market. With the increasing adoption of electric vehicles, the significance of efficient battery charging methods like CC-CV has become more pronounced. As lithium-ion batteries are sensitive to voltage and temperature variations, a well-implemented CC-CV charging strategy allows for better thermal management and minimizes the risk of battery degradation [3]. Furthermore, advancements in this charging technique aim to reduce charging times, a critical factor influencing consumer acceptance of electric vehicles [4]. By optimizing the CC-CV charging process, manufacturers can significantly enhance battery performance metrics such as

longevity, capacity retention, and safety, thereby reinforcing the overall sustainability of electric mobility solutions.

II. PRINCIPLE OF CC-CV CHARGING

Constant current constant voltage (CC-CV) charging is a charging method employed for lithium-ion batteries in electric vehicles (EVs), combining the principles of both constant current (CC) and constant voltage (CV) charging phases. During the CC phase, a constant charging current is supplied to the battery, ensuring a steady and controlled flow of electrical energy (I_c) [5]. This phase allows for rapid charging of the battery, maximizing charging efficiency and reducing charging time. As the battery reaches a predetermined voltage threshold, the charging process transitions to the CV phase. In this phase, the charging voltage (V_c) across the battery terminals is held constant, preventing overcharging and ensuring the battery's safety and longevity. The CV phase allows the battery to absorb the remaining charge at a controlled voltage level, minimizing the risk of overvoltage and thermal runaway. The Fig.1 illustrating the CC-CV charging process would depict a graph with time on the x-axis and voltage/current on the y-axis. Initially, during the CC phase, the charging current remains constant while the voltage gradually increases. Once the voltage reaches the CV threshold, the charging voltage is held constant, and the charging current decreases as the battery becomes fully charged. This dual-phase charging strategy optimizes the charging process, balancing rapid charging with careful voltage control to maximize battery performance and lifespan.

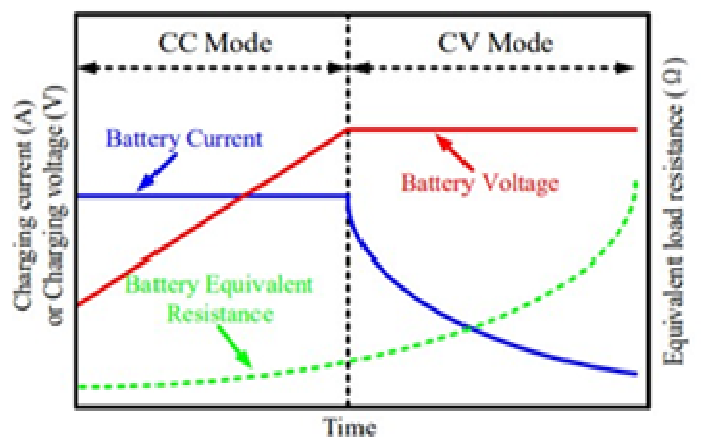


Fig.1. The CC-CV charging scheme showing the two main phases [6].

III. MODELLING OF CC-CV CHARGING

A. Constant Current Modelling;

During the CC phase, the charger provides a constant current I_{cc} and the battery voltage $V(t)$ gradually rises as the battery's state of charge (SOC) increases.

For a initial condition let $V(0)$ be the initial voltage. The SOC is low, and the charger operates in constant current mode, supplying a steady current $E_{total} = Ecc + Ecv$.

The relationship between current, voltage, and SOC can be described using the following differential equation:

$$I_{cc} = C \frac{d(SOC)}{dt} \quad (1)$$

Where, I_{cc} = Charging current in Amp

C = Charging capacity in Ah

SOC= State of charge

Since the voltage of the battery is dependent on the SOC, we can model the voltage rise during the CC phase as:

$$V(t) = V_0 + I_{cc} \cdot R_{int} + f(SOC) \quad (2)$$

Where, R_{int} = Internal resistance of the battery in Ohms

$f(SOC)$ = Voltage-SOC relationship, which can be obtained empirically or approximated using polynomial fits.

The duration of the CC phase t_{cc} can be determined by the time it takes for the voltage to reach the maximum allowed voltage V_{max}

$$t_{cc} = \frac{C \cdot (SOC_{max} - SOC_0)}{I_{cc}} \quad (3)$$

B. Constant Voltage Modelling

When the battery voltage V_{max} , reaches the charger switches to the CV phase, maintaining a constant voltage while the current decreases exponentially.

The charging current during the CV phase is modeled as:

$$I(t) = I_0 \cdot e^{-\frac{t}{RC}} \quad (4)$$

Where, I_0 = Current at the beginning of the CV phase

R = Equivalent resistance of the battery

C = Battery capacity

In this phase, the current decreases as the SOC approaches full capacity. The total charge added to the battery during the CV phase is:

$$Q_{CV} = \int_0^{t_{CV}} I(t) dt = I_0 \cdot R \cdot C \cdot (1 - e^{-\frac{t_{CV}}{RC}}) \quad (5)$$

C. Constant Current-Constant Voltage Modelling

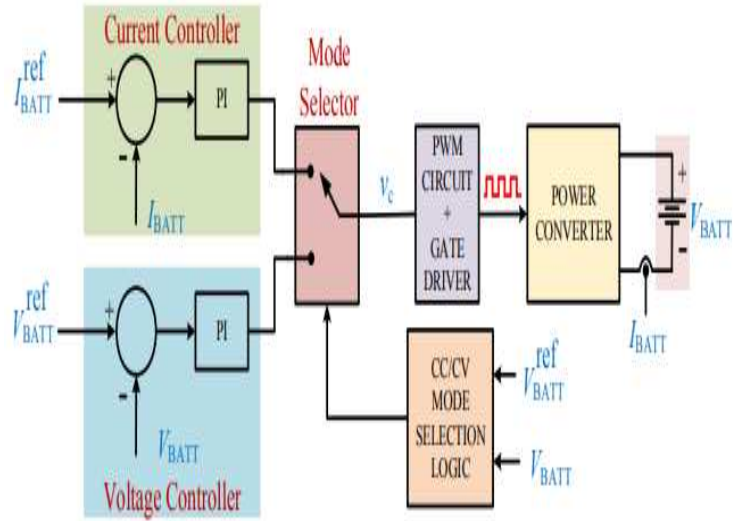
The complete charging process is modeled by combining the CC and CV phases. The total energy delivered to the battery during the CC phase is

$$E_{cc} = I_{cc} \cdot V(t) \cdot t_{cc} \quad (6)$$

For the CV phase, the energy is:

$$E_{CV} = \int_0^{t_{CV}} I(t) dt = V_{max} \cdot I_0 \cdot R \cdot C \cdot (1 - e^{-\frac{t_{CV}}{RC}}) \quad (7)$$

The total energy delivered to the battery during the charging



cycle is the sum of the two phases:

$$E_{total} = Ecc + Ecv \quad (8)$$

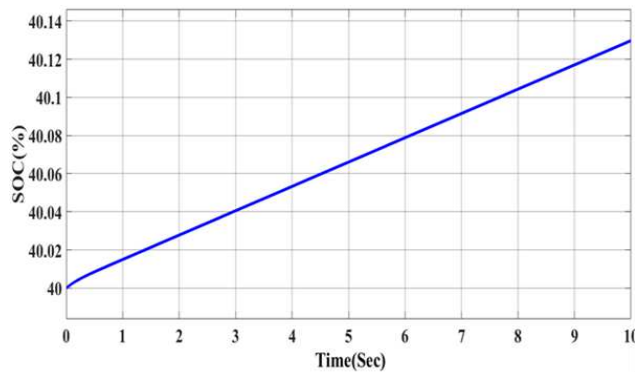
Fig.2. Control block diagram of a conventional CC-CV Battery Charge [7].

IV. SIMULATION RESULTS

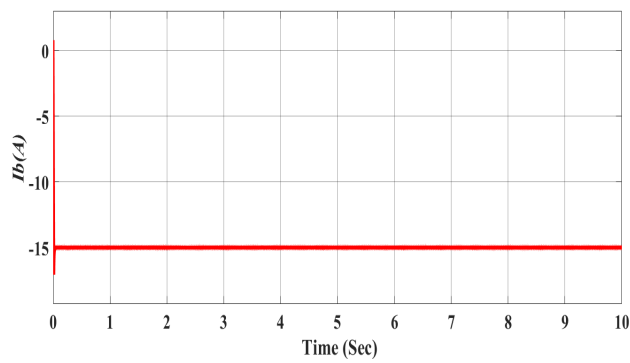
A. Constant Current(CC) Charging;

In analyzing the simulation results of the constant current (CC) charging mode, several key observations emerge from the Fig.3 depicting state of charge (SoC), current, and voltage profiles. Firstly, In Fig.3 (a) the SoC curve illustrates a gradual increase over time, indicating the charging process effectively replenishing the battery's energy storage. Concurrently, in Fig.3 (b) the current graph demonstrates a consistent flow during the CC phase, validating the mode's ability to maintain a steady charging rate. This stability is crucial for optimizing both battery

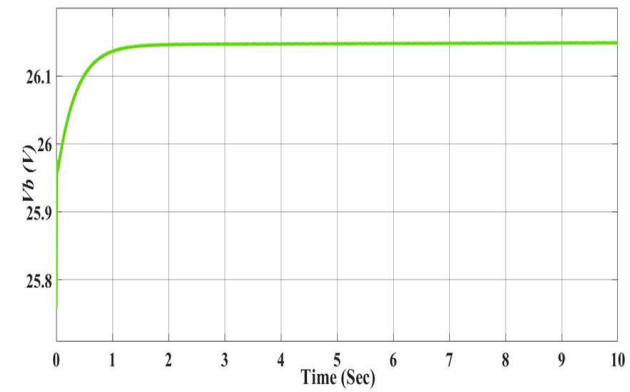
lifespan and charging efficiency. Moreover, the voltage profile in Fig.3 (c) showcases an anticipated rise as the battery becomes increasingly charged, reaching its peak towards the end of the CC phase.



(a)



(b)



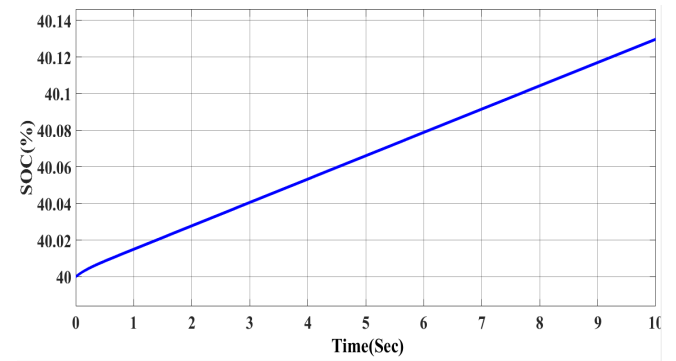
(c)

Fig.3 Responses of (a) SOC (b) battery current (c) battery voltage in CC charging mode.

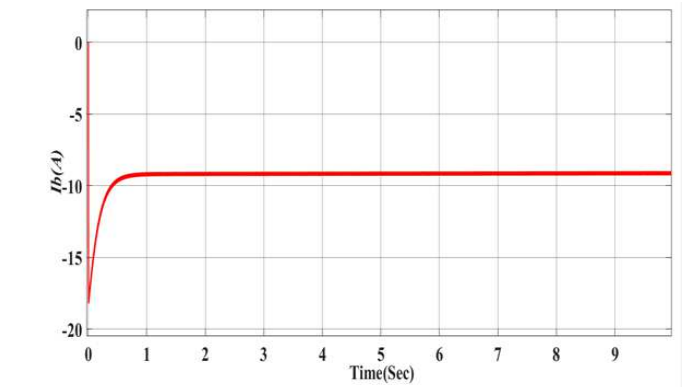
B. Constant Voltage (CV) Charging:

In analyzing the simulation results of the constant voltage (CV) charging mode, several key observations emerge from the Fig.4 depicting state of charge (SoC), current, and voltage profiles. Firstly, In Fig.4 (a) the SoC curve illustrates a gradual increase over time, indicating the charging process effectively replenishing the battery's

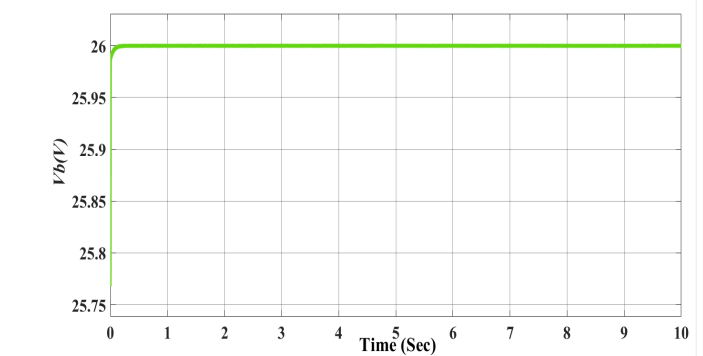
energy storage. Concurrently, in Fig. 4 (b) the current graph demonstrates a gradually increase during the CV phase. The voltage profile depicted in Fig. 4 (c) remains stable throughout the charging process, underscoring the CV mode's capability to regulate voltage and prevent overcharging



(a)



(b)



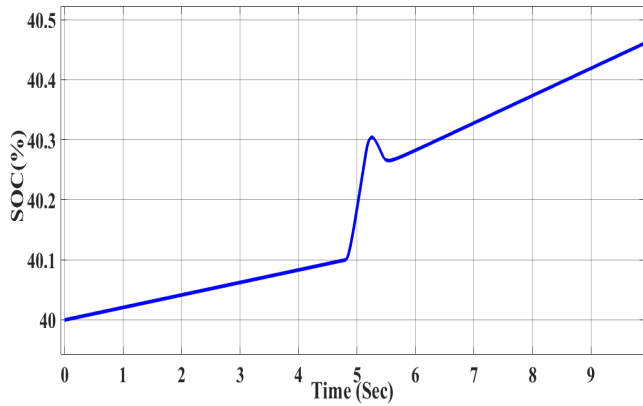
(c)

Fig.4 Responses of (a) SOC (b) battery current (c) battery voltage in CV charging mode.

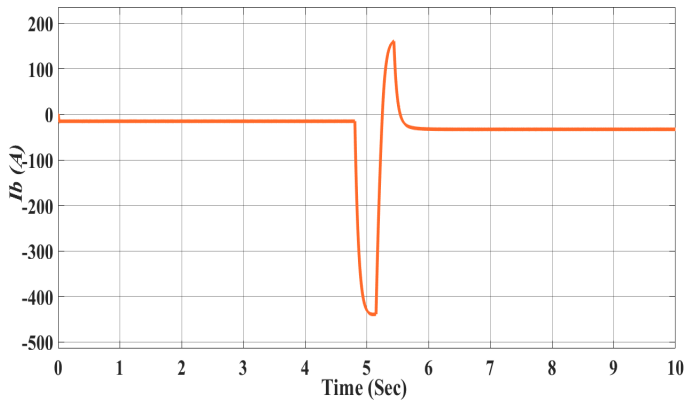
C. Constant Current Constant Voltage (CC-CV) Charging):

In this method, the state of charge (SoC) battery is set to 40.1 %. When the battery SoC reaches to 40.1%, the voltage of battery attains its predefined maximum value and the operation mode of battery charger is switched from CC mode to CV mode. Then the charging current drops to

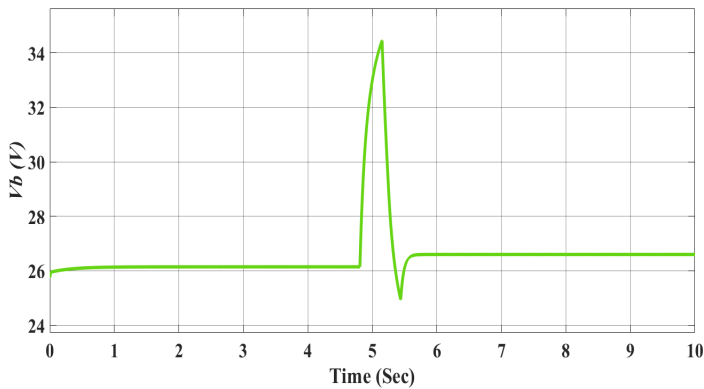
prevent of overcharged and overstressing the battery voltage. In Fig.5 (a)-(c), the results of CC-CV method are presented.



(a)



(b)



(c)

Fig.5. Responses of (a) SOC (b) battery current (c) battery voltage in CC-CV charging mod

V. CONCLUSION

The constant current constant voltage (CC-CV) charging represents a robust and efficient method for replenishing energy in lithium-ion batteries for electric vehicles. By combining the benefits of controlled current and voltage phases, CC-CV charging optimizes charging efficiency while safeguarding battery health. Continued research and optimization of CC-CV charging protocols are crucial for advancing the performance and sustainability of electric transportation systems.

APPENDIX

- I. Lithium-ion Battery
 Nominal voltage =24V
 Rated Capacity =20 Ah
 SOC=40 %

REFERENCES

- [1] D. Purushothaman, R Narayanamoorthi, et al, "Charging ahead: Unlocking the potential of constant voltage and constant current modes in WPT for EVs", Journal of Energy Storage, Vol.96, Aug 2024.
- [2] G.H. Chen, C.L. Lew et al. "Implementation of Constant Temperature-Constant Voltage Charging Method with Energy Loss Minimization for Lithium-Ion Batteries", Electronics., Vol.13, No.3, Feb, 2024.
- [3] D. Shuangming, K. Xia, et al, "Optimization charging method of lithium-ion battery based on multi-objective BBO algorithm", Journal of Energy Storage, Vol.91, June, 2024.
- [4] V Marupuru, D. Kishan, "Efficient and cost-effective wireless CC/CV charging for electric vehicles: A bipolar duty cycle approach", Computers and Electrical Engineering, Vol.118, Aug, 2024.
- [5] M. Brenna et al., "Electric Vehicles Charging Technology Review and Optimal Size Estimation", Journal of Electrical Engineering & Technology, Vol.15, pp.2539-2552, Oct 2020.
- [6] S.H. Huang et al., "Reconfigurable Hybrid Resonant Topology for Constant Current/Voltage Wireless Power Transfer of Electric Vehicles", Electronics, Vol.9, No.8, Aug 2020.
- [7] S. M. Ferdous, M. Asaduzzaman Shueb, G. Shafiullah and M. A. Moin Oninda, "Parallel Resonant Converter for Battery Charging Application", 2019 9th International Conference on Power and Energy Systems (ICPES), Perth, WA, Australia, 2019, pp. 1-6.
- [8] G. Chen and W. H. Chung, "Evaluation of Charging Methods for Lithium-Ion Batteries", Electronics, vol. 12, no. 4095, 2023.
- [9] L. R. Dung and J. H. Yen, "ILP-based Algorithm for Lithium-Ion Battery Charging Profile", in Proc. 2010 IEEE Int. Symp. on Industrial Electronics, Bari, Italy, 2010, pp. 1-6.
- [10] J. M. Amanor-Boadu, A. Guiseppi-Elie, and E. Sánchez-Sinencio, "Search for Optimal Pulse Charging Parameters for Li-Ion Polymer Batteries Using Taguchi Orthogonal Arrays", IEEE Trans. Ind. Electron., vol. 65, no. 9, pp. 8982–8992, Sep. 2018.